MATH 477: Applied Finite Element Methods

Assignment

December 1, 2024

NAZARBAYEV
UNIVERSITY

Submitted for **MATH 477: Applied Finite Element Method** at Department of Mathematics, the School of Sciences and Humanities, Nazarbayev University

Student Name: Alen Abdildayev

- Alen Abdildayev - ID: 201995353

Course Project
Instructor Name: **Dongming Wei**

In submitting this work we are indicating that we have read the University's Academic Integrity Policy. We declare that all material in this assessment is our own work except where there is clear acknowledgement and reference to the work of others.

Problems

1D Problem

a. Describe a 1D physical problem.

b. Solve it by Laplace Transform, and derive the exact solution.

c. Derive finite element solution, the local and global finite element matrices, and apply boundary conditions.

d. Plot two solutions together in MATLAB.

2D Problem

a. Describe a 2D physical problem.

b. Derive two-element finite element solution, the two local and global finite element matrices, and apply boundary conditions.

c. Plot of the two element solution in MATLAB.

d. Plot 2D Problem in COMSOL and compare with two-element matrix solution.

3D Problem

a. Present PDE of 3D Model, the boundary/initial conditions.

b. Explain each coefficient as material constants, their physical meanings, the boundary conditions, geometric parameters etc.

c. Plot 3D Model in COMSOL.

Solutions

1D Problem

a. Describe a 1D physical problem.

First we simplify our 3D Problem into a 1D. In one dimension, our room will consist of two boundary conditions, and it will have length 10 meters. On one hand we have heat loss at 268.15 K, on the other hand we have heat gain at 298.15 K. For our model, we will use next appropriate equation - Steady-State Heat Transfer Equation:

$$-k\frac{d^2u}{dx^2} = Q_{gen};$$

where $-k\frac{d^2u}{dx^2}$ models heat conduction in our room, and Q_{gen} the amount of heat generated in our room.

Then, we describe boundary conditions (both are Robin Conditions) for our system:

At $x = 0$:
$-ku'(0) = h_1 \cdot (-5 - u(0))$ - Convective Heat Loss at one end (windows) ;

At $x = 10$:
$-ku'(0) = h_2 \cdot (u(10) - 25)$ - Convective Heat Gain at the other end (open door) ;

where, k - thermal conductivity of our material, and h_1, h_2 - heat transfer coefficient. We use 5^0C and 25^0C instead of Kelvins, for simplicity.

Since there is no heat source in our room, that means Q_{gen} is equal to 0. The reason why we used this equation, is because it's a steady-state, thus a second derivative of heat will be equal to 0. Because in our problem we assume a wind from the open door to window, let's assume that $h_1 = h_2 = 10\frac{W}{m^2 \cdot K}$, and for simplicity $k = 1$. Thus, our final equation for 1D will be:

$$u''(x) = 0; \quad -u'(0) = 10 \cdot (-5 - u(0)); \quad -u'(10) = 10 \cdot (u(10) - 25); \quad 0 \le x \le 10;$$

b. Solve it by Laplace Transform, and derive the exact solution.

$$-u'' = 0 \Rightarrow \mathcal{L}(-u'') = 0 \Rightarrow s^2 \cdot U(s) - su(0) - u'(0) = 0 \Rightarrow U(s) = u(0) + \frac{u'(0)}{s};$$

$$u(x) = u(0) + x \cdot u'(0) \Rightarrow u'(x) = u'(0) = u'(10) \Rightarrow -50 - 10 \cdot u(0) = 10 \cdot u(10) - 250;$$

$$u(x) = u(0) + xu'(0) \Rightarrow u(10) = u(0) + 100 \cdot u(0) + 500; u(10) = 20 - u(0);$$

$$u(0) = -\frac{480}{102}; u'(0) = -\frac{4800}{102} + 50;$$

Thus:

$$u_{exact} = \left(-\frac{4800}{102} + 50\right) \cdot x - \frac{480}{102};$$

c. Write two finite element solution, the local and global finite element matrices, and apply boundary conditions.

If we don't know the exact solution of the problem, we could use Galerkin's Finite Element Equation. To use him, let our global nodes be: $x_1 = 0; x_2 = \frac{10}{3}; x_3 = \frac{20}{3}; x_4 = 10$.

Then, we need to find nodal values u_i for $i = 1, 2, 3, 4$ by calculating matrices $K^{(e)}, F^{(e)}$, for $e = 1, 2$. But before that, we need to find $p(x), q(x), f(x)$. To find them, we use next equation:

$$-\frac{d}{dx}\left(p(x)\frac{du(x)}{dx}\right) + q(x)u(x) = f(x);$$

Because our equation is $u''(x) = 0$,that means $f(x) = 0$; $q(x) = 0$ and $p(x) = -1$. Therefore:

For linear we have $[x_1, x_2] = [0, \frac{10}{3}], l^{(1)} = \frac{10}{3}, p^{(1)} = 1, q^{(1)} = 0, f^{(1)} = 0$. Thus:

$$K^{(1)} = \frac{p^{(1)}}{l^{(1)}}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} + \frac{q^{(1)}l^{(1)}}{6}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} \frac{3}{10} & -\frac{3}{10} \\ -\frac{3}{10} & \frac{3}{10} \end{bmatrix};$$

$$F^{(1)} = \frac{f^{(1)}l^{(1)}}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix};$$

For quadratic we have $[x_2, x_4] = [\frac{10}{3}, 10], l^{(2)} = \frac{20}{3}, p^{(2)} = 1, q^{(2)} = 0, f^{(2)} = 0$. Thus:

$$K^{(2)} = \frac{p^{(2)}}{3l^{(2)}}\begin{bmatrix} 7 & -8 & 1 \\ -8 & 16 & -8 \\ 1 & -8 & 7 \end{bmatrix} + \frac{q^{(2)}l^{(2)}}{30}\begin{bmatrix} 4 & 2 & -1 \\ 2 & 16 & 2 \\ -1 & 2 & 4 \end{bmatrix} = \begin{bmatrix} \frac{7}{20} & -\frac{4}{10} & \frac{1}{20} \\ -\frac{4}{10} & \frac{8}{10} & -\frac{4}{10} \\ \frac{1}{20} & -\frac{4}{10} & \frac{7}{20} \end{bmatrix};$$

$$F^{(2)} = \frac{f^{(2)}l^{(2)}}{6}\begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix};$$

Then we compute global stiffness matrix, global load vector, and additional load vector:

$$K = \begin{bmatrix} K_{11}^{(1)} & K_{12}^{(1)} & 0 & 0 \\ K_{21}^{(1)} & K_{22}^{(1)} + K_{11}^{(2)} & K_{12}^{(2)} & K_{13}^{(2)} \\ 0 & K_{21}^{(2)} & K_{22}^{(2)} & K_{23}^{(2)} \\ 0 & K_{31}^{(2)} & K_{32}^{(2)} & K_{33}^{(2)} \end{bmatrix} = \begin{bmatrix} \frac{3}{10} & -\frac{3}{10} & 0 & 0 \\ -\frac{3}{10} & \frac{13}{20} & -\frac{4}{10} & \frac{1}{20} \\ 0 & -\frac{4}{10} & \frac{8}{10} & -\frac{4}{10} \\ 0 & \frac{1}{20} & -\frac{4}{10} & \frac{7}{20} \end{bmatrix}$$

$$F = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}; Q = \begin{bmatrix} 10u_1 + 50 \\ 0 \\ 0 \\ 250 - 10u_4 \end{bmatrix}; \Rightarrow Ku = F + Q;$$

Then by using boundary conditions we get our final system:

$$\begin{bmatrix} \frac{3}{10} & -\frac{3}{10} & 0 & 0 \\ -\frac{3}{10} & \frac{13}{20} & -\frac{4}{10} & \frac{1}{20} \\ 0 & -\frac{4}{10} & \frac{8}{10} & -\frac{4}{10} \\ 0 & \frac{1}{20} & -\frac{4}{10} & \frac{7}{20} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} 10u_1 + 50 \\ 0 \\ 0 \\ 250 - 10u_4 \end{bmatrix}$$

By solving set of equations, we get our answers:

$$u_1 \approx -5.299; u_2 = 4.6924; u_3 = 14.6846; u_4 = 24.6768;$$

For local linear element, we have these shape functions:

$$N_1^{(1)}(x) = \frac{x_2 - x}{x_2 - x_1}; N_2^{(1)}(x) = \frac{x - x_1}{x_2 - x_1};$$

For quadratic element, we have these shape functions:

$$N_2^{(2)}(x) = \frac{(x - x_3)(x - x_4)}{(x_2 - x_3)(x_2 - x_4)}; N_3^{(2)}(x) = \frac{(x - x_2)(x - x_4)}{(x_3 - x_2)(x_3 - x_4)}; N_4^{(2)} = \frac{(x - x_2)(x - x_3)}{(x_4 - x_2)(x_4 - x_3)};$$

Since our global nodes are $x_1 = 0; x_2 = \frac{10}{3}; x_3 = \frac{20}{3}; x_4 = 10$, our shape functions will be equal to:

$$N_1^{(1)}(x) = 1 - \frac{3x}{10};$$

$$N_2^{(1)}(x) = \frac{3x}{10};$$

$$N_2^{(2)}(x) = \frac{3}{2}\left(\frac{3x}{10} - 2\right)\left(\frac{x}{10} - 1\right);$$

$$N_3^{(2)}(x) = -3\left(\frac{3x}{10} - 1\right)\left(\frac{x}{10} - 1\right);$$

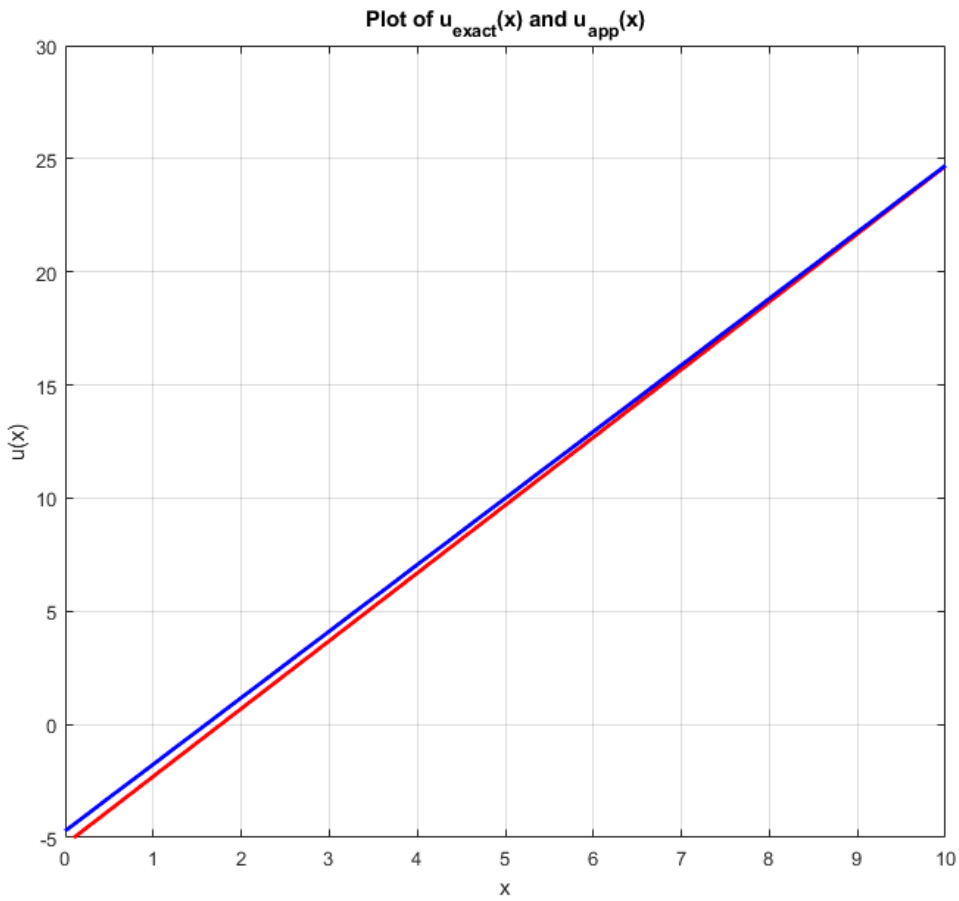
$$N_4^{(2)}(x) = \frac{1}{2}\left(\frac{3x}{10} - 1\right)\left(\frac{3x}{10} - 2\right);$$

Thus for $u_{app}(x) = \sum_{i=1}^{4} N_j(x)u_j$, our final answer is:

$$u_{app}(x) = \begin{cases} 2.99742x - 5.299, & 0 \le x \le \frac{10}{3} \\ 2.99766x - 5.2998, & \frac{10}{3} \le x \le 1 \end{cases};$$

Since our problem is $-u'' = 0$, it make sense why our u_{app} is linear, and why we get almost the same linear function from linear element and quadratic element.

d. Plot two solutions together in MATLAB.



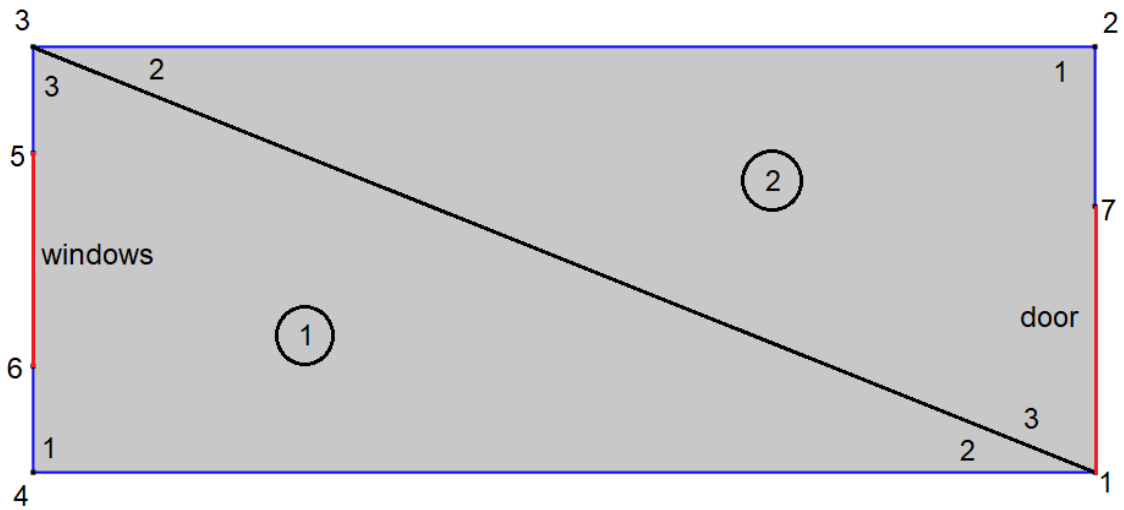
In the plot above, the blue function is the u_{exact}, while red function is u_{app}. It's evident, that u_{app} is practically the same as u_{exact}, which means we used Galerkin's Finite Element Equation correctly.

2D Problem

a. Describe a 2D physical problem.

Compared to 1D, in 2D we get an additional dimension - height. That means, that oour boundary conditions will be slightly adjusted. Therefore, our equation with boundary conditions for 2D will be:

$$\begin{cases} -\nabla^2 u(x,y) = 0, \text{in } \Omega = (0,10) \times (0,4); \\ -\frac{\partial u}{\partial n} = 10(-5 - u(x,y)), \text{on } \Gamma_{56} = \{0\} \times [1,3]; \\ -\frac{\partial u}{\partial n} = 10(u(x,y) - 25), \text{on } \Gamma_{17} = \{10\} \times [0,2.5]; \\ \frac{\partial u}{\partial n} = 0, \text{on } \Gamma_{46} \cup \Gamma_{41} \cup \Gamma_{27} \cup \Gamma_{23} \cup \Gamma_{35} = (0,10) \times \{0,4\} \cup \\ \cup (0,10) \times \{0,4\} \cup \{0\} \ \times (0,1) \cup \{0\} \ \times (3,4) \cup \{10\} \ \times (2.5,4); \end{cases}$$



b. Derive two-element finite element solution, the two local and global finite element matrices, and apply boundary conditions.

First, we need to establish some general equations for our solution. For any triangles, with three points $(x_1^{(e)}, y_1^{(e)}), (x_2^{(e)}, y_2^{(e)}), (x_3^{(e)}, y_3^{(e)})$, we have these values:

$$a_1^{(e)} = \frac{y_2^{(e)} - y_3^{(e)}}{|\Omega^{(e)}|}; a_2^{(e)} = -\frac{y_1^{(e)} - y_3^{(e)}}{|\Omega^{(e)}|}; a_3^{(e)} = \frac{y_1^{(e)} - y_2^{(e)}}{|\Omega^{(e)}|};$$

$$b_1^{(e)} = -\frac{x_2^{(e)} - x_3^{(e)}}{|\Omega^{(e)}|}; b_2^{(e)} = \frac{x_1^{(e)} - x_3^{(e)}}{|\Omega^{(e)}|}; b_3^{(e)} = -\frac{x_1^{(e)} - x_2^{(e)}}{|\Omega^{(e)}|};$$

From here our stiffness matrix and force vector for such triangle is:

$$K^{(e)} = k|\Omega^{(e)}| \begin{bmatrix} a_1^{(e)}a_1^{(e)} + b_1^{(e)}b_1^{(e)} & a_1^{(e)}a_2^{(e)} + b_1^{(e)}b_2^{(e)} & a_1^{(e)}a_3^{(e)} + b_1^{(e)}b_3^{(e)} \\ a_2^{(e)}a_1^{(e)} + b_2^{(e)}b_1^{(e)} & a_2^{(e)}a_2^{(e)} + b_2^{(e)}b_2^{(e)} & a_2^{(e)}a_3^{(e)} + b_2^{(e)}b_3^{(e)} \\ a_3^{(e)}a_1^{(e)} + b_3^{(e)}b_1^{(e)} & a_3^{(e)}a_2^{(e)} + b_3^{(e)}b_2^{(e)} & a_3^{(e)}a_3^{(e)} + b_3^{(e)}b_3^{(e)} \end{bmatrix}; F^{(e)} = \frac{f|\Omega^{(e)}|}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix};$$

Since in Ω, $-\nabla^2 u(x,y) = 0$, it means our $f = 0$, thus we can not include $F^{(e)}$. However, we have convective heat transfers, and we should add them.

$$K_b^{(e)} = \frac{\beta_{12}^{(e)} l_{12}^{(e)}}{6} \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \frac{\beta_{23}^{(e)} l_{23}^{(e)}}{6} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} + \frac{\beta_{31}^{(e)} l_{31}^{(e)}}{6} \begin{bmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 2 \end{bmatrix};$$

$$f_b^{(e)} = \frac{\beta_{12}^{(e)} T_\infty^{12} l_{12}^{(e)}}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \frac{\beta_{23}^{(e)} T_\infty^{23} l_{23}^{(e)}}{2} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \frac{\beta_{31}^{(e)} T_\infty^{31} l_{31}^{(e)}}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix};$$

Now, we can start solving our problem.

$$NE = 2, NL = 3, NG = 4 \Rightarrow C(NE, NL) = \begin{bmatrix} 4 & 1 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

For lower triangle we have:

$$k^{(1)} = 1, \quad f^{(1)} = 0, \quad (x_1^{(1)}, y_1^{(1)}) = (0,0), \quad (x_2^{(1)}, y_2^{(1)}) = (10,0), \quad (x_3^{(1)}, y_3^{(1)}) = (0,4);$$

$$|\Omega^{(1)}| = 20, \quad l_{12}^{(1)} = 10, \quad l_{23}^{(1)} = 2\sqrt{29}, \quad l_{31}^{(1)} = 4, \quad \beta_{31}^{(1)} = 10, \quad T_\infty^{31} = 268.15;$$

$$a_1^{(1)} = \frac{y_2^{(1)} - y_3^{(1)}}{|\Omega^{(1)}|} = -\frac{1}{5}; a_2^{(1)} = -\frac{y_1^{(1)} - y_3^{(1)}}{|\Omega^{(1)}|} = \frac{1}{5}; a_3^{(1)} = \frac{y_1^{(1)} - y_2^{(1)}}{|\Omega^{(1)}|} = 0;$$

$$b_1^{(1)} = -\frac{x_2^{(1)} - x_3^{(1)}}{|\Omega^{(1)}|} - \frac{1}{2}; b_2^{(1)} = \frac{x_1^{(1)} - x_3^{(1)}}{|\Omega^{(1)}|} = 0; b_3^{(1)} = -\frac{x_1^{(1)} - x_2^{(1)}}{|\Omega^{(1)}|} = \frac{1}{2};$$

From these values, out stiffness matrix and convective heat flux is:

$$K^{(1)} = 20 \begin{bmatrix} \frac{29}{100} & -\frac{1}{25} & -\frac{1}{4} \\ -\frac{1}{25} & \frac{1}{25} & 0 \\ -\frac{1}{4} & 0 & \frac{1}{4} \end{bmatrix}, \quad f_b^{(1)} = \frac{10 \cdot 268.15 \cdot 2}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad K_b^{(1)} = \frac{10 \cdot 2}{6} \begin{bmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 2 \end{bmatrix};$$

For upper triangle we have:

$$k^{(2)} = 1, \quad f^{(2)} = 0, \quad (x_1^{(2)}, y_1^{(2)}) = (10,4), \quad (x_2^{(2)}, y_2^{(2)}) = (0,4), \quad (x_3^{(2)}, y_3^{(2)}) = (10,0);$$

$$|\Omega^{(2)}| = 20, \quad l_{12}^{(2)} = 10, \quad l_{23}^{(2)} = 2\sqrt{29}, \quad l_{31}^{(2)} = 4, \quad \beta_{31}^{(2)} = 10, \quad T_\infty^{31} = 298.15;$$

$$a_1^{(2)} = \frac{y_2^{(2)} - y_3^{(2)}}{|\Omega^{(2)}|} = \frac{1}{5}; a_2^{(2)} = -\frac{y_1^{(2)} - y_3^{(2)}}{|\Omega^{(2)}|} = -\frac{1}{5}; a_3^{(2)} = \frac{y_1^{(2)} - y_2^{(2)}}{|\Omega^{(2)}|} = 0;$$

$$b_1^{(2)} = -\frac{x_2^{(2)} - x_3^{(2)}}{|\Omega^{(2)}|}\frac{1}{2}; b_2^{(2)} = \frac{x_1^{(2)} - x_3^{(2)}}{|\Omega^{(2)}|} = 0; b_3^{(2)} = -\frac{x_1^{(2)} - x_2^{(2)}}{|\Omega^{(2)}|} = -\frac{1}{2};$$

From these values, out stiffness matrix and convective heat flux is:

$$K^{(2)} = 20 \begin{bmatrix} \frac{29}{100} & -\frac{1}{25} & -\frac{1}{4} \\ -\frac{1}{25} & \frac{1}{25} & 0 \\ -\frac{1}{4} & 0 & \frac{1}{4} \end{bmatrix}, \quad f_b^{(2)} = \frac{10 \cdot 298.15 \cdot 2.5}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad K_b^{(2)} = \frac{10 \cdot 2.5}{6} \begin{bmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 2 \end{bmatrix};$$

By combining both triangles, we get our global stiffness matrix and force vector:

$$K = 20\begin{bmatrix} \frac{1}{25} & 0 & 0 & -\frac{1}{25} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{25} & 0 & -\frac{1}{4} & \frac{29}{100} \end{bmatrix} + 20\begin{bmatrix} \frac{1}{4} & -\frac{1}{4} & 0 & 0 \\ -\frac{1}{4} & \frac{29}{100} & -\frac{1}{25} & 0 \\ 0 & -\frac{1}{25} & \frac{1}{25} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \frac{10 \cdot 2.5}{6}\begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} +$$

$$+\frac{10 \cdot 2}{6}\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} \frac{212}{15} & -\frac{5}{6} & 0 & -\frac{4}{5} \\ -\frac{5}{6} & \frac{212}{15} & -\frac{4}{5} & 0 \\ 0 & -\frac{4}{5} & \frac{187}{15} & -\frac{5}{3} \\ -\frac{4}{5} & 0 & -\frac{5}{3} & \frac{187}{15} \end{bmatrix};$$

$$F = \frac{10 \cdot 298.15 \cdot 2.5}{2}\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \frac{10 \cdot 268.15 \cdot 2}{2}\begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{29815}{8} \\ \frac{29815}{8} \\ \frac{5363}{2} \\ \frac{5363}{2} \end{bmatrix};$$

Thus, our final system is:

$$\begin{bmatrix} \frac{212}{15} & -\frac{5}{6} & 0 & -\frac{4}{5} \\ -\frac{5}{6} & \frac{212}{15} & -\frac{4}{5} & 0 \\ 0 & -\frac{4}{5} & \frac{187}{15} & -\frac{5}{3} \\ -\frac{4}{5} & 0 & -\frac{5}{3} & \frac{187}{15} \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} \frac{29815}{8} \\ \frac{29815}{8} \\ \frac{5363}{2} \\ \frac{5363}{2} \end{bmatrix};$$

If we solve it, we get our approximate values:

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} 296.47167 \\ 296.47167 \\ 270.24790 \\ 270.24790 \end{bmatrix};$$

Now we need to find shape functions for lower and upper triangle. We will find them by using next equations:

$$a_1^{(e)} = \frac{x_2y_3 - x_3y_2}{2|\Omega^{(e)}|}; a_2^{(e)} = \frac{x_3y_1 - x_1y_3}{2|\Omega^{(e)}|}; a_3^{(e)} = \frac{x_1y_2 - x_2y_1}{2|\Omega^{(e)}|};$$

$$b_1^{(e)} = \frac{y_2 - y_3}{2|\Omega^{(e)}|}; b_2^{(e)} = \frac{y_3 - y_1}{2|\Omega^{(e)}|}; b_3^{(e)} = \frac{y_1 - y_2}{2|\Omega^{(e)}|};$$

$$c_1^{(e)} = \frac{x_3 - x_2}{2|\Omega^{(e)}|}; c_2^{(e)} = \frac{x_1 - x_3}{2|\Omega^{(e)}|}; c_3^{(e)} = \frac{x_2 - x_1}{2|\Omega^{(e)}|};$$

$$N_i^{(e)} = a_i^{(e)} + b_i^{(e)}x + c_i^{(e)}y;$$

For lower triangle shape functions are:

$$a_1^{(1)} = \frac{x_2y_3 - x_3y_2}{2|\Omega^{(1)}|} = 1; a_2^{(1)} = \frac{x_3y_1 - x_1y_3}{2|\Omega^{(1)}|} = 0; a_3^{(1)} = \frac{x_1y_2 - x_2y_1}{2|\Omega^{(1)}|} = 0;$$

$$b_1^{(1)} = \frac{y_2 - y_3}{2|\Omega^{(1)}|} = -\frac{1}{10}; b_2^{(1)} = \frac{y_3 - y_1}{2|\Omega^{(1)}|} = \frac{1}{10}; b_3^{(1)} = \frac{y_1 - y_2}{2|\Omega^{(1)}|} = 0;$$

$$c_1^{(1)} = \frac{x_3 - x_2}{2|\Omega^{(1)}|} = -\frac{1}{4}; c_2^{(1)} = \frac{x_1 - x_3}{2|\Omega^{(1)}|} = 0; c_3^{(1)} = \frac{x_2 - x_1}{2|\Omega^{(1)}|} = \frac{1}{4};$$

Thus:

$$N_1^{(1)}(x,y) = 1 - \frac{x}{10} - \frac{y}{4}; \quad N_2^{(1)}(x,y) = \frac{x}{10}; \quad N_3^{(1)}(x,y) = \frac{y}{4};$$

For upper triangle shape functions are:

$$a_1^{(2)} = \frac{x_2y_3 - x_3y_2}{2|\Omega^{(2)}|} = -1; a_2^{(2)} = \frac{x_3y_1 - x_1y_3}{2|\Omega^{(2)}|} = 1; a_3^{(2)} = \frac{x_1y_2 - x_2y_1}{2|\Omega^{(2)}|} = 1;$$

$$b_1^{(2)} = \frac{y_2 - y_3}{2|\Omega^{(2)}|} = \frac{1}{10}; b_2^{(2)} = \frac{y_3 - y_1}{2|\Omega^{(2)}|} = -\frac{1}{10}; b_3^{(2)} = \frac{y_1 - y_2}{2|\Omega^{(2)}|} = 0;$$

$$c_1^{(2)} = \frac{x_3 - x_2}{2|\Omega^{(2)}|} = \frac{1}{4}; c_2^{(2)} = \frac{x_1 - x_3}{2|\Omega^{(2)}|} = 0; c_3^{(2)} = \frac{x_2 - x_1}{2|\Omega^{(2)}|} = -\frac{1}{4};$$

Thus:

$$N_1^{(2)}(x,y) = \frac{x}{10} + \frac{y}{4} - 1; \quad N_2^{(2)}(x,y) = 1 - \frac{x}{10}; \quad N_3^{(2)}(x,y) = 1 - \frac{y}{4};$$

Then we have characteristic set functions:

$$\psi^{(1)}(x,y) = \begin{cases} 1, \text{if} & |\Omega^{(1)}|; \\ 0, & |\Omega^{(2)}|; \end{cases}$$

$$\psi^{(2)}(x,y) = \begin{cases} 1, \text{if} & |\Omega^{(2)}|; \\ 0, & |\Omega^{(1)}|; \end{cases}$$

Thus we get these global shape functions:

$$N_1(x,y) = N_2^{(1)}(x,y)\psi^{(1)}(x,y) + N_3^{(2)}(x,y)\psi^{(2)}(x,y) = \frac{x}{10}\psi^{(1)}(x,y) + \left(1 - \frac{y}{4}\right)\psi^{(2)}(x,y);$$

$$N_2(x,y) = N_1^{(2)}(x,y)\psi^{(2)}(x,y) = \left(\frac{x}{10} + \frac{y}{4} - 1\right)\psi^{(2)}(x,y);$$

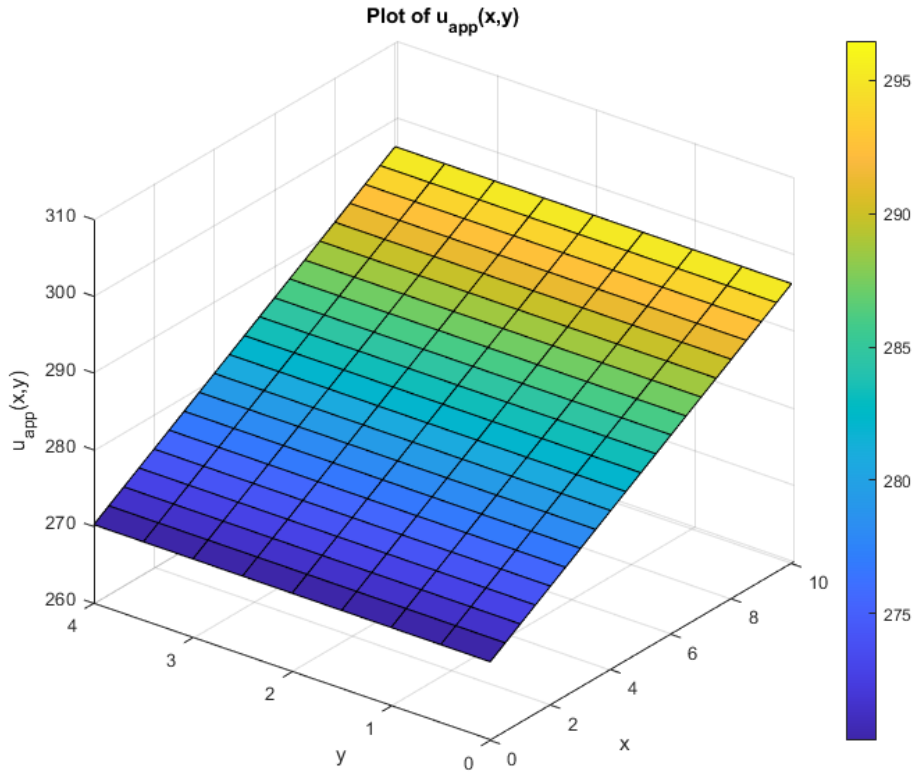
$$N_3(x,y) = N_3^{(1)}(x,y)\psi^{(1)}(x,y) + N_2^{(2)}(x,y)\psi^{(2)}(x,y) = \frac{y}{4}\psi^{(1)}(x,y) + \left(1 - \frac{x}{10}\right)\psi^{(2)}(x,y);$$

$$N_4(x,y) = N_1^{(1)}(x,y)\psi^{(1)}(x,y) = \left(1 - \frac{x}{10} - \frac{y}{4}\right)\psi^{(1)}(x,y);$$

Since $u_{app}(x,y) = \sum_{i=1}^{4} N_j(x,y)u_j$, we write our solution:

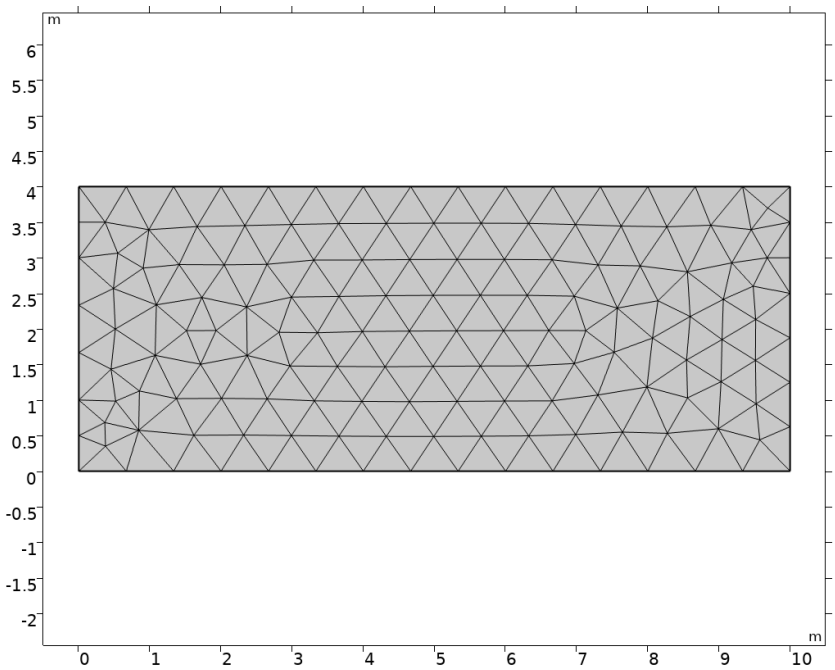
$$u_{app}(x,y) = \begin{cases} 296.47167 \cdot \frac{x}{10} + 270.24790 \cdot \frac{y}{4} + 270.24790 \cdot (1 - \frac{x}{10} - \frac{y}{4}), & |\Omega^{(1)}|; \\ 296.47167 \cdot (1 - \frac{y}{4}) + 296.47167 \cdot (\frac{x}{10} + \frac{y}{4} - 1) + 270.24790 \cdot (1 - \frac{x}{10}), & |\Omega^{(2)}|; \end{cases}$$

c. Plot of the two element solution in MATLAB.

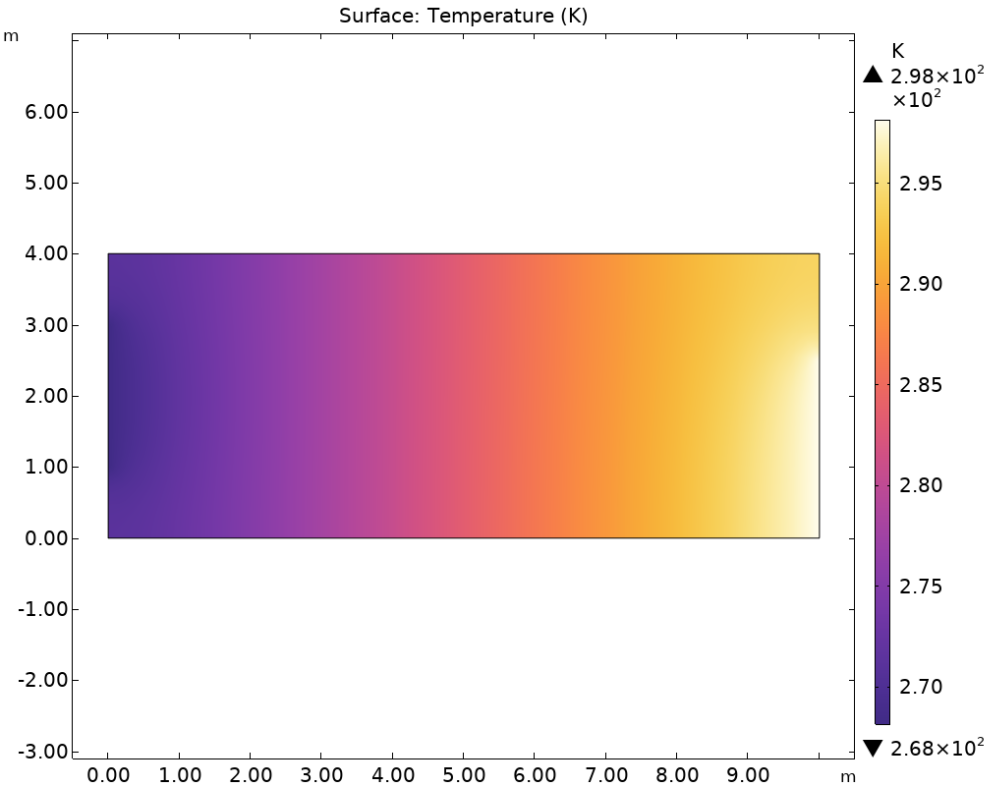


Above you can see plot of the u_{app}. Since we used only two triangles, it's make sense why it's linear.

d. Plot 2D Problem in COMSOL and compare with two-element matrix solution.

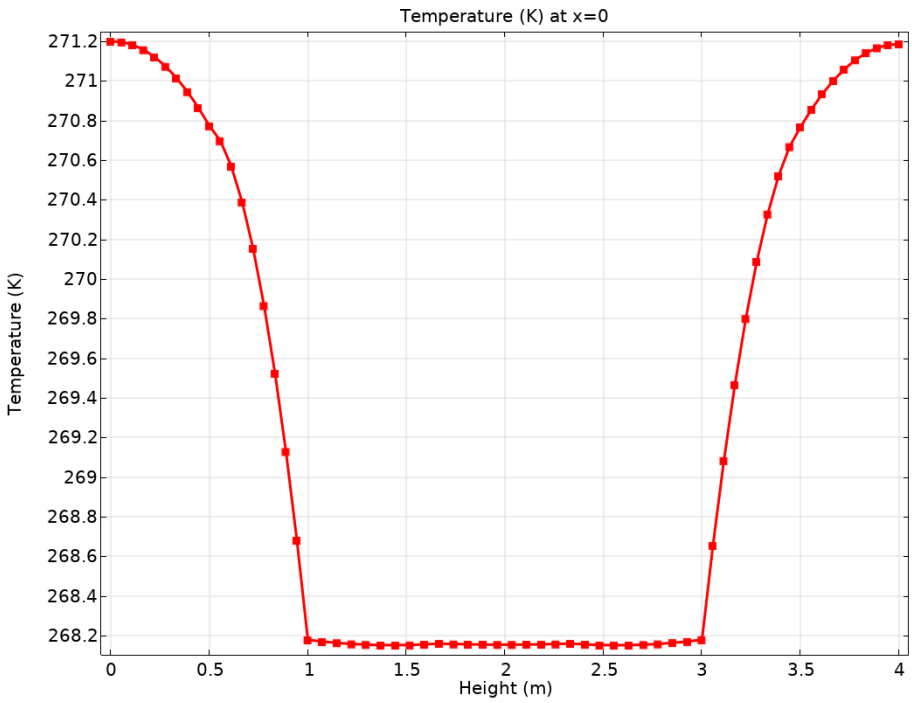


In the picture above, you can see mesh for this 2D problem.

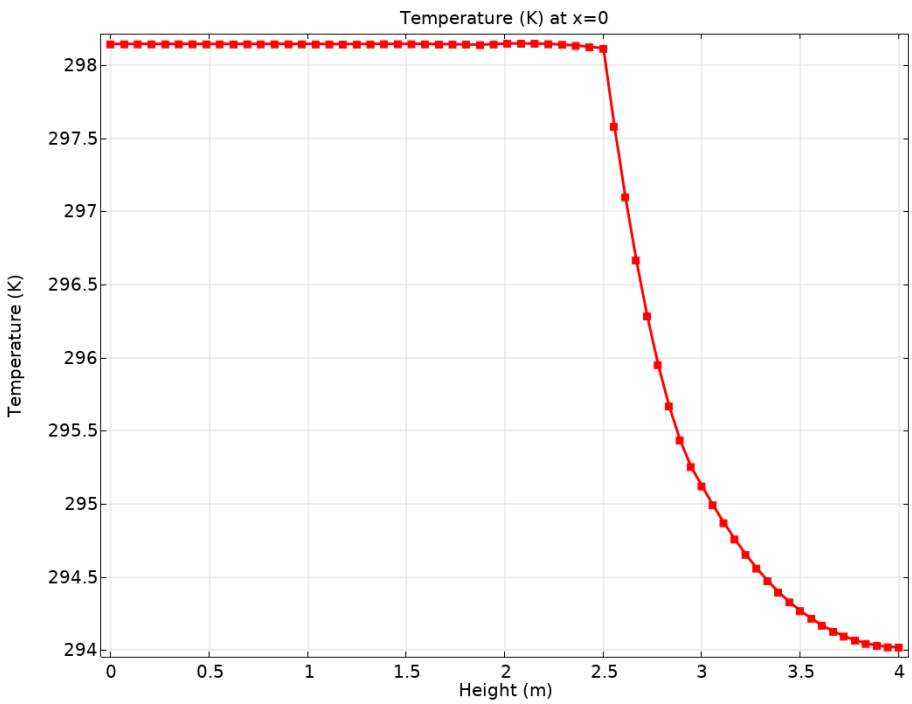


In the picture above, you can see 2D temperature distribution in our room. It's clearly see from the right side, that there is a temperature difference in height. It's

also exist from the window side, however it is not so noticeable at all. Therefore, let plot 1D Temperature/Distance graph for three scenarios: Windows side, Door side and Floor Side.

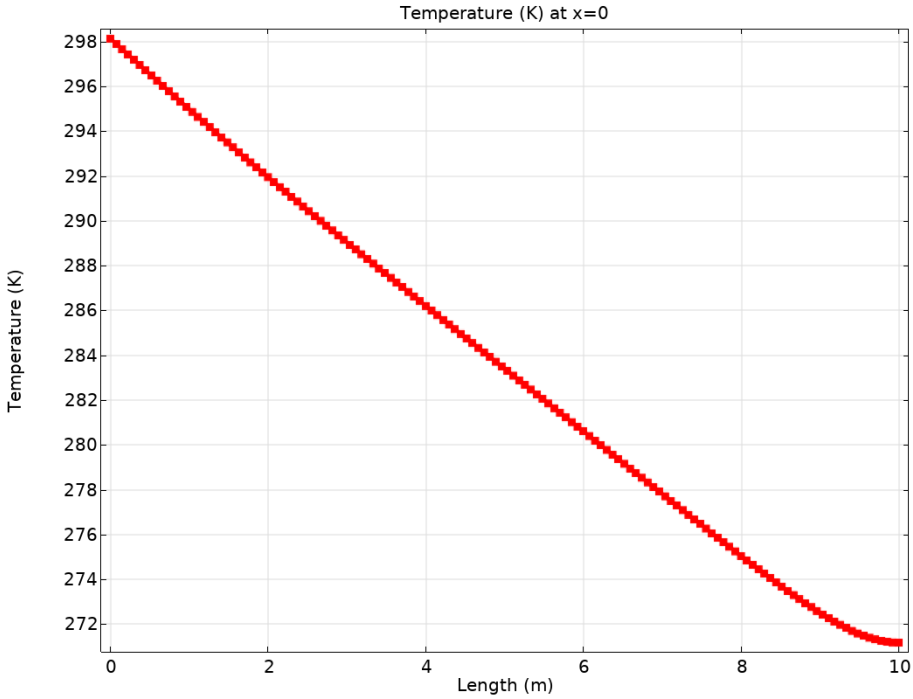


In the first picture you can see 1D plot on windows side. The temperature varies from $\approx 268.2K$ to $\approx 271.2K$, which effectively gives us a $\approx 3K$ difference.



In the next picture you see 1D plot on door side. Here, temperature varies from

$\approx 298K$, and decreases to $\approx 294K$, which gives us a $\approx 4K$ difference. With this information we can conclude, that there is not much difference between window and door side in temperature distribution over height. Perhaps, because of the choice of color for the plot, we have a false illusion.



Lastly, we see the 1D plot on floor side. As expected, we see here a linear decrease in temperature from left side to the right side.

Now, if we compare COMSOL plot and MATLAB plot, we can clearly see the differences. First of all, we see the temperature variation relatively to height, while in MATLAB the temperature does not depend on height. This is because COMSOL uses many more triangles compared to our direct calculations. However, horizontally, the MATLAB plot is very close to the COMSOL plot, indicating that we have done a good calculation.

3D Problem

a. Present PDE of 3D Model, the boundary/initial conditions

Now we finally reached 3D, and we can add another dimension - thickness. Since there is no need to calculate finite element solution, we can use real values of the air, and then solve it in the COMSOL. Thus, our problem is a heat transfer in a rectangular room. It has two walls which are parallel to each other, two windows from one side and door from the other. On the other side of the door, it is warm - 25 degrees Celsius, while on the other side of the window, it is cold - 5 degrees Celsius.

Since it is a room, our material is air. Here are the parameters of the room, door and window.

Name	Expression	Value	Description
T_door	298.15[K]	298.15 K	Hot Temperature
T_windows	268.15[K]	268.15 K	Cold Temperature
h_air	10[W/(m^2*K)]	10 W/(m²·K)	Heat Transfer Coefficient...
lenx	10[m]	10 m	Length of Room [x]
leny	6[m]	6 m	Length of Room [y]
height	4[m]	4 m	Height of Room [z]

For our 3D Model, I used COMSOL - Heat Transfer in Solids and Stationary. Thus my heat transfer equation is:

$$\rho C_\rho u \cdot \nabla T + \nabla \cdot q = Q + Q_{ted}; q = -k\nabla T;$$

b. Explain each coefficient as material constants, their physical meanings, the boundary conditions, geometric parameters etc.

In our new model, we got new material constants. Let's get through them.
$\rho = 1.225\ kg/m^3$ - density of the air in the room ;

$C_\rho = 1005\ J/(kg \cdot K)$ - specific heat capacity of air at constant pressure ;

u - velocity vector ;

T - temperature of our system and ∇T - temperature gradient ;

q - heat flux vector, Q - heat source term and Q_{ted} - thermal energy density term;

$k = 0.02624\ W/(m \cdot K)$ - thermal conductivity of the material ;

In 3D Problem our boundary conditions would be next:

$$\begin{cases} \rho C_\rho u \cdot \nabla T + \nabla \cdot q = Q + Q_{ted}; q = -k\nabla T, \text{in Room ;} \\ -\frac{\partial u}{\partial n} = 10(-5 - u(x,y)), \text{on Windows ;} \\ -\frac{\partial u}{\partial n} = 2(u(x,y) - 25), \text{on Doors ;} \\ \frac{\partial u}{\partial n} = 0, \text{on Walls, Ceiling, Floor ;} \end{cases}$$

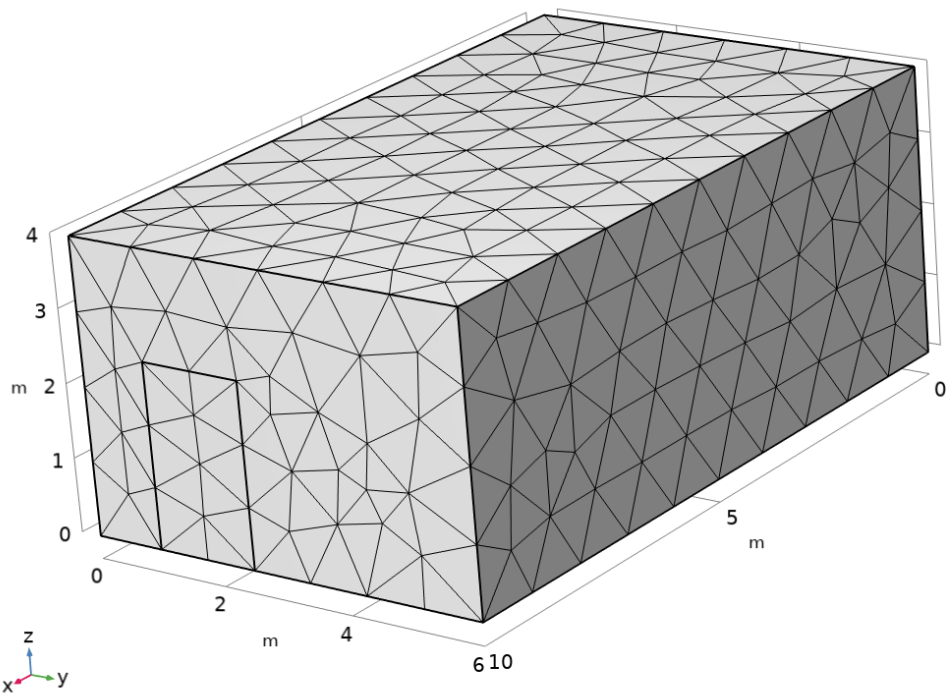
Geometric parameters will be next:

Surface area from (0,1,1) to (0,2.5,3) and from (0,3.5,1) to (0,5,3) is windows.

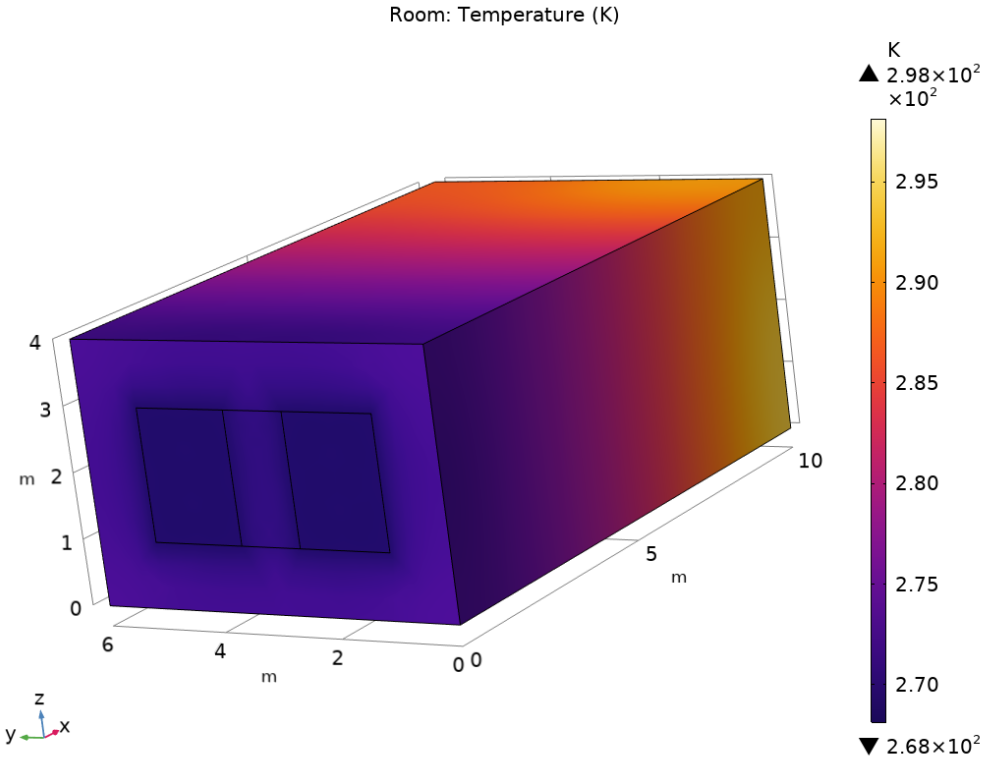
Surface area from (10,1,0) to (10,2.5,2.5) is doors.

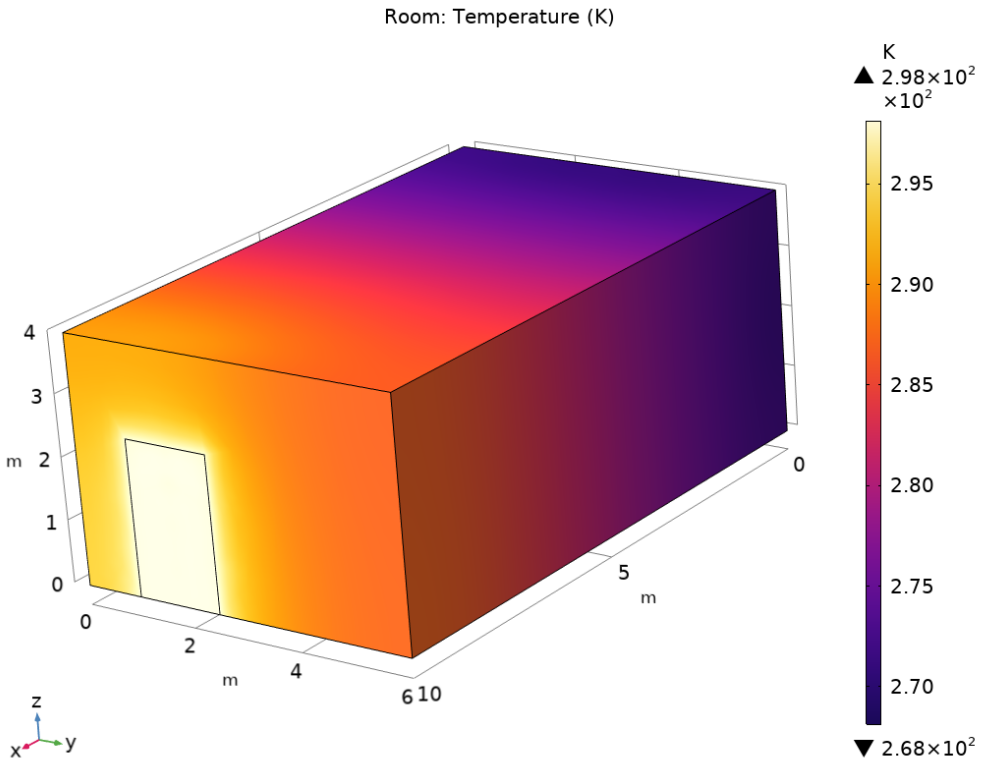
Anything else is walls, ceiling or floor.

c. Plot 3D Model in COMSOL.

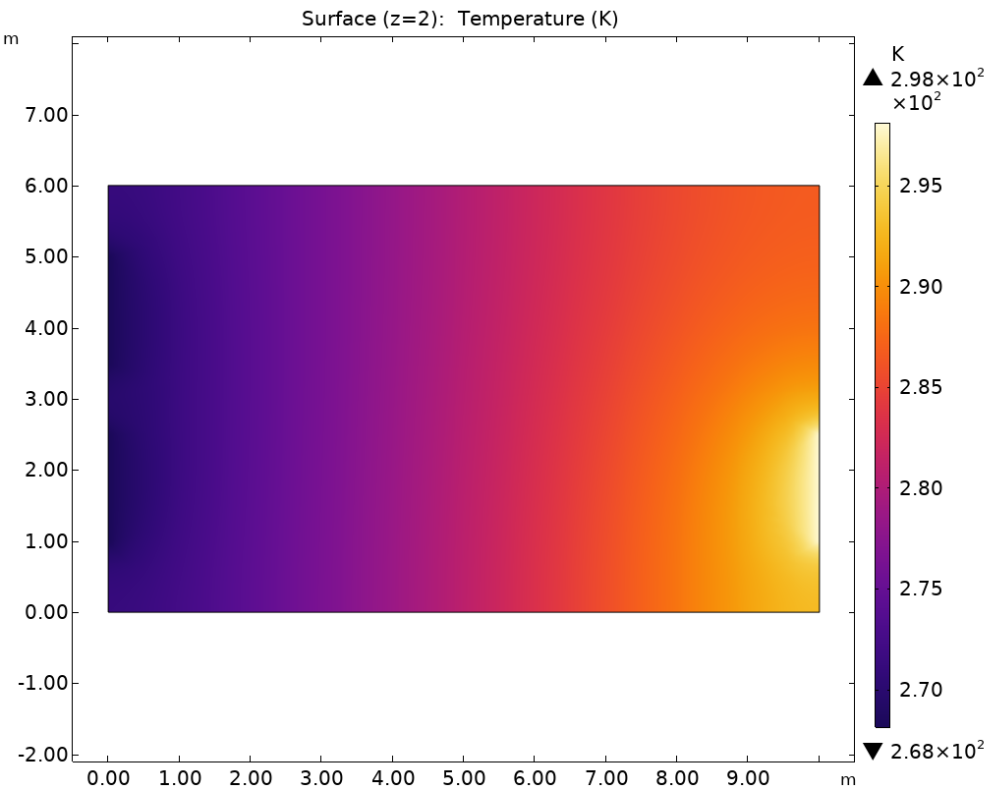


In the picture above, you can see mesh for this 3D problem.


Room: Temperature (K)
K
▲ 2.98×10²
×10²
2.95
2.90
2.85
2.80
2.75
2.70
▼ 2.68×10²
4
3
m 2
1
0
6
4
2
0
m
0
5
m
10
z
y
x


Room: Temperature (K)
K
▲ 2.98×10²
×10²
2.95
2.90
2.85
2.80
2.75
2.70
▼ 2.68×10²
4
3
m 2
1
0
0
2
m
4
6
10
0
5
m
z
x
y

In two pictures above, you can see temperature distribution in our room from two different angles. From the first angle you can see windows, from the second angle you can see door. It's worth noting that area between windows have higher temperature than temperature close to the windows. On the other hand, since door is not in the center of the wall, we could assume that the less value of y we have, the higher temperature we have. Let's prove that



In the picture above you can see slice of our 3D room at height 2 meters. Even though the difference in temperature for y values is small, it's still exist.

Discussion

Choosing Problem

When our group met first time each other, we decided that studying heat flux would be the best idea. I specifically decided to study it with the conditions of room with windows and door.

The reason for that was because it reminds me of our room on this course, where we had lectures. And sometimes in that room, windows where open. And because of that, it became interesting for me, what temperature will be in the room after a long. long time.

Motivation to Solve It

However, an idea alone wasn't enough for the motivation. Another reason I wanted to solve this problem was because of my interest in understanding heat equations. Ever since I took a partial differential equations course a year ago, I've wanted to understand them. This course provided a perfect opportunity to do so, by using COMSOL or solving them by hand through finite element methods or numerical solutions.

Additionally, I was motivated to learn COMSOL. I had heard of the program before, however I never felt the urge to use it. But this problem gave me the reason to learn it, therefore I now kinda understand COMSOL.

Real Life Applications

As for real-life applications, I think it would be useful to understand what will be the steady-state temperature inside a room after a long time if we open or close doors and windows. Since COMSOL allows us to model any type of room, it means we can solve the problem for various room configurations.

Appendix

MATLAB Code for 1D Problem, part d:

```
x=linspace(0,10,1000);

% Define functions u_app(x), u_exact(x)
uleft=(2.99742*x-5.299).*(x<10/3);
uright=(2.99766*x-5.2998).*(10/3<=x);
u_approximate=uleft+uright;
u_exact=(-4800/102+50)*x-480/102;

% Plot u_app(x), u_exact(x)
plot(x,u_approximate,'r','linewidth',2')
hold on
plot(x,u_exact,'b','linewidth',2')
xlabel('x'); ylabel('u(x)'), grid(); xlim([0 10]); ylim([-5 30]);
title('Plot of u_{exact}(x) and u_{app}(x)');
```

MATLAB Code for 2D Problem, part c:

```
[x, y]=meshgrid(0:0.5:10);

% Define functions u_app(x,y)
ulower=(270.24790*(1-0.1*x-0.25*y)+296.47167*0.1*x+270.24790*0.25*y).*((0.1*x+0.25*y)<1);
uupper=(296.47167*(1-0.25*y)+270.24790*(1-0.1*x)+296.47167*(0.1*x+0.25*y-1)).*(1<=(0.1*x+0.25*y));
u_app=ulower+uupper;

% Plot functions u_app(x,y)
grid(); surf(x,y,u_app);
xlabel('x'); ylabel('y'); zlabel('u_{app}(x,y)');
xlim([0 10.1]); ylim([0 4]); zlim([260 310]);
title('Plot of u_{app}(x,y)'); colorbar;
```